

Hw6

1a) $\sum_{i=1}^n f_i^2 = f_n \cdot f_{n+1}$

$$f_1^2 + f_2^2 + \dots + f_n^2 = f_n \cdot f_{n+1}$$
$$f_1 = 1, f_2 = 1$$

Induction:

Base case ($n=1$): $f_1^2 = 1 = 1 \cdot 1 = f_1 \cdot f_2$

Inductive Step: Assume true for $n=k$, then $n=k+1$:

$$\sum_{i=1}^{k+1} f_i^2 = f_k \cdot f_{k+1} + f_{k+1}^2 = f_{k+1} (f_k + f_{k+1}) = f_{k+1} f_{k+2}$$

1b) $x_{n+1} = x_n + y_n$

$$y_{n+1} = x_n$$

$$y_0 = 0, x_0 = 1$$

Prove that $y_n = f_n$ (n -th Fibonacci number):

$$n=0: x_0 = 1, y_0 = 0$$

$$n=1: x_1 = x_0 + y_0 = 1 + 0 = 1, y_1 = x_0 = 1$$

$$n=2: x_2 = x_1 + y_1 = 1 + 1 = 2, y_2 = x_1 = 1$$

$$n=3: x_3 = x_2 + y_2 = 2 + 1 = 3, y_3 = x_2 = 2$$

Matches

Fibonacci
numbers

$y_0 = 0, y_1 = 1, y_2 = 1, y_3 = 2$, By induction: Assume $y_k = f_k$ and $y_{k-1} = f_{k-1}$, then:

$$y_{k+1} = x_k = x_{k-1} + y_{k-1} = y_k + y_{k-1} = f_k + f_{k-1} = f_{k+1}$$